INTRODUCTION

Inborn errors of metabolism (IEM) or inherited metabolic disorders are a group of disorders with specific enzyme defects that impede with normal metabolism of protein, fat or carbohydrate. As its name implies, inborn errors mean birthdefects in young infants which are inherited. IEM like galactosemia, phenylketonuria, alkaptonuria, albinism, Tay -Sachs disease and von Gierke disease, can appear atbirth or later in life .

A food product that is not broken down into energy can build up in the body and cause a wide range of symptoms. Several inborn errors of metabolism cause developmental delays or other medical problems if they are not controlled. The main indication of IEM is an excess storage or accumulation of specific metabolites in tissues, organs and blood which further manifest to health diseases. Most IEM are rare but some are life threatening.

The metabolism of our body comprises two major balanced activities: anabolism (synthesis) and catabolism (degradation). The absence or deficiency of an enzyme will cause an abnormal accumulation of the intermediate products of metabolism in the body and increased excretion in urine as such, or, as their degradation products. Some of the intermediates could even be toxic. For example, in the following reaction

$$R \xrightarrow{a} B \xrightarrow{b} C \xrightarrow{c} D \xrightarrow{d} P$$

- R is the reactant,
- B, C and D are intermediates, P is the product
- a, b, c and d are enzymes catalyzing the indicated steps of the reactions.
- In this pathway, if any enzyme is deficient or absent, the previous intermediate accumulates and produces toxicity. It also affects the amount of product (P) formed and thereby leads to a disease.

The classification of IEM is presented in the following table

A few important diseases associated with IEM are given below:

Disorders	Examples
Disorders of aminoacid metabolism	Phenylketonuria, Alkaptonuria, Albinism Homocysteinuria
Disorders of carbohydrate metabolism	Galactosemia, hereditary fructose intolerance, Glycogen storage disorders (Von Gierke disease)
Lysosomal storage disorders	Mucopolysaccharidosis, Tay-Sachs disease, Niemann-Pick disease, Gaucher's disease
Organic acidemia	Methyl malonic acidemia

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